Tab 1

Fork

Tab 2

Planetary Pulse

Planetary Pulse is an experimental interface that converts Earth system data into sound and visual signals.

The goal is to allow humans to intuitively perceive planetary stability and ecological imbalance.

Instead of reading complex datasets, listeners can hear the state of the Earth as evolving harmonies, rhythms, and distortions.

Concept

Human cognition evolved to detect patterns in sound and motion, not large numerical systems.

By mapping planetary indicators to musical parameters we can create an interface where instability becomes audible.

Example mappings:

CO_2 concentration → pitch
Ocean temperature → bass intensity
Biodiversity index → number of active instruments
Ecosystem stress → distortion

Stable Earth = harmonic soundscape
Destabilized Earth = dissonant system signal

Long-Term Vision

This prototype explores a future planetary interface where humanity can monitor Earth's systems in real time.

Possible evolution:

1. Planetary Pulse (sonification)
2. Earth Dashboard (AI digital twin visualization)
3. Planetary Autopilot (AI-assisted equilibrium systems)

Inspired by the Kardashev Scale concept of civilization evolution.

Tab 3

```
import numpy as np
import sounddevice as sd

# Example planetary indicators
co2 = 420
ocean_temp = 1.2
biodiversity = 0.75

# Map data to sound
pitch = 200 + (co2 - 350)
bass = 60 + ocean_temp * 40
instrument_count = int(biodiversity * 10)

duration = 3
sample_rate = 44100
t = np.linspace(0, duration, int(sample_rate * duration))

signal = np.sin(2 * np.pi * pitch * t)

sd.play(signal, sample_rate)
sd.wait()
```

Tab 4